Ridge-Cal

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Ridge-Cal: Efficient Regularized Calibration of External Prognostic Scores Using Blinded Trial Data

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Abstract

Background. External prognostic scores improve randomized trial efficiency via covariate adjustment (PROCOVA). However, when the trial population differs from the historical data used to build the score, the score may be miscalibrated, reducing or reversing the efficiency gain.

Methods. We propose Ridge-Cal, a two-step procedure that (1) diagnoses miscalibration by comparing predictive accuracy of the score alone versus the score plus a small set of pre-specified calibration covariates, and (2) recalibrates the score via ridge-penalized Cox regression on the trial's blinded data. The ridge penalty — which controls the strength of recalibration — is selected automatically by cross-validation within the trial.

Results. In simulations under severe population shift, Ridge-Cal recovers 7.5 percentage points of power over standard PROCOVA (0.758 vs 0.833), reduces bias by 80% (0.035 vs 0.007), and maintains nominal Type I error (0.052). When no shift is present, the power penalty is minimal ($-$0.8 percentage points) and the diagnostic correctly indicates no recalibration is needed.

Conclusion. Ridge-Cal provides a framework for recalibrating external prognostic scores using only blinded trial data. It works with any black-box score, requires no unblinding, and selects its regularization strength automatically via cross-validation within the trial.

Keywords: PROCOVA; covariate adjustment; prognostic score; ridge regression; LoRA; model fine-tuning

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1. Introduction

1.1 Covariate Adjustment in Randomized Trials

Covariate adjustment in randomized clinical trials improves statistical power by accounting for baseline prognostic factors. For continuous outcomes, ANCOVA is the standard approach; for time-to-event endpoints, the Cox proportional hazards model with covariate adjustment achieves similar gains (Tsiatis, 2006; Hajage et al., 2018). Regulatory guidance (FDA, 2023; EMA, 2015) emphasizes pre-specification and parsimony, typically restricting adjustment to a handful of stratification variables.

1.2 PROCOVA and the Problem of Population Shift

PROCOVA (Schuler et al., 2022) addresses the parsimony constraint by using an external dataset to build a rich prognostic model that compresses multi-dimensional baseline data into a single score. When used as a covariate in the primary analysis, this score captures information from many covariates while maintaining a parsimonious model. The EMA qualified PROCOVA for phase II/III trials (EMA, 2022), and FDA guidance has acknowledged the approach (FDA, 2024). For time-to-event endpoints, the analogous approach uses the Cox PH model with the prognostic score as a single covariate (Hajage et al., 2018).

A key assumption of PROCOVA is that the external model is well-calibrated for the trial population. This assumption fails when the external and trial populations differ — a common scenario in oncology, where eligibility criteria evolve, standard of care changes, and biomarker assays improve over time. When the score is miscalibrated, the estimated treatment effect can be biased and the efficiency gain reduced.

1.3 Existing Approaches to Population Shift

Several approaches exist to handle population shift:

- **Bayesian dynamic borrowing** uses power priors (Ibrahim & Chen, 2000), commensurate priors (Hobbs et al., 2011; 2012), or meta-analytic predictive priors (Schmidli et al., 2014) to down-weight historical control data when it conflicts with the current trial. These methods borrow patient-level data; they do not update the prognostic model itself.
- **Domain adaptation** methods (Pan & Yang, 2010) adjust predictive models for covariate shift but are rarely applied to survival outcomes in clinical trials.
- **Liao et al. (2025)** proposed using external prognostic scores in doubly-robust estimators, but without a calibration step for population shift.

All of these approaches treat the prognostic score as fixed. None update the score using trial data.

1.4 A New Perspective: Model Fine-Tuning

We draw inspiration from Low-Rank Adaptation (LoRA; Hu et al., 2022), a parameter-efficient fine-tuning method for large language models. LoRA freezes a pre-trained model's weights and learns a small, regularized update using a small domain-specific dataset. The update is constrained to be low-rank, preventing overfitting and catastrophic forgetting.

We apply this same principle to prognostic score calibration:

Concept	LLM Fine-Tuning (LoRA)	Ridge-Cal (This Paper)
Pre-trained model	LLM on large corpus	External score from historical data
Fine-tuning data	Small domain dataset	Trial data (blinded, all patients)
Update structure	Low-rank matrices $\Delta W = BA$	Ridge-penalized coefficients for $\mathcal{C}$
Regularization	Rank r controls update size	Ridge λ controls update size
Hyperparameter selection	Holdout validation	K-fold cross-validation within trial
Catastrophic forgetting prevention	Low-rank constraint	Shrinkage toward $\beta_j = 0$

The analogy highlights a key insight: **calibration of a prognostic score should be efficient and regularized, not a full refit.** The external score already captures the bulk of the prognostic information. Only a small, targeted subset of coefficients may need updating, and the update should be regularized to prevent overfitting when the trial sample size is limited.

1.5 Contributions

1. **Diagnostic framework.** A simple test — compare the C-index of the score alone versus the score plus calibration covariates on blinded trial data — determines whether recalibration is needed.
2. **Regularized recalibration (Ridge-Cal).** A ridge-penalized Cox model on blinded trial data that learns a calibration correction for a pre-specified subset of covariates, with the penalty strength selected automatically by cross-validation.

3. **Simulation evidence.** In 10,000-rep simulations under severe population shift, Ridge-Cal recovers 7.5 percentage points of power over standard PROCOVA with nominal Type I error and a 0.8 pp penalty when no shift is present.

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2. Method

2.1 Notation

Consider a randomized trial with N patients. For patient i, we observe baseline covariates $W_i \in \mathbb{R}^p$, treatment assignment $A_i \in \{0, 1\}$ (randomized 1:1), and a time-to-event outcome (T_i, δ_i) where $\delta_i = 1$ indicates the event of interest. An external dataset of N_{ext} patients provides historical training data.

An external prognostic model has already been fit on the external data and produces a score $\hat{S}_i^{(ext)} = f(W_i)$ for any input W_i. The function f may be any predictive model — a Cox PH, random survival forest, SuperLearner, gradient boosting machine, or neural network. We treat f as a black box; we do not require access to its internal parameters.

The p baseline covariates are partitioned into two sets: - $\mathcal{C}$: **calibration covariates** ($c \ll p$), a small set expected a priori to be susceptible to population shift. These are pre-specified in the statistical analysis plan, based on clinical judgment (e.g., acute-phase reactants like CRP and albumin, biomarkers whose relevance varies by disease setting, demographic factors with known cross-trial variability). - $\mathcal{F}$: **fixed covariates** ($p - c$), the remainder. The external model's handling of these covariates is assumed to be adequate.

2.2 Diagnostic Step

We first determine whether recalibration is needed. Using the trial's **blinded data** (all patients, ignoring treatment assignment), we fit two Cox proportional hazards models:

Model 1 (base):

$$\lambda_1(t \mid W_i) = \lambda_{01}(t) \exp(\beta_0 + \beta_1 \hat{S}_i^{(ext)})$$

Model 2 (augmented):

$$\lambda_2(t \mid W_i) = \lambda_{02}(t) \exp(\beta_0 + \beta_1 \hat{S}_i^{(ext)} + \beta_{\mathcal{C}}^T W_{i,\mathcal{C}})$$

The concordance indices C_1 and C_2 are compared. If $C_2 - C_1 > \delta$ (we use $\delta = 0.01$ throughout), the score is diagnosed as miscalibrated with respect to $\mathcal{C}$, and the fine-tuning step is triggered. This diagnostic uses only blinded data and requires no unblinding.

Comparison to LoRA. In LLM fine-tuning, the need for adaptation is assessed by evaluating the pre-trained model's performance on a domain-specific validation set. Our C-index comparison plays the same role — it tests whether the pre-trained score ("base model") is adequate or requires fine-tuning.

2.3 Fine-Tuning (Ridge-Cal)

When fine-tuning is indicated, we fit a ridge-penalized Cox model (Friedman et al., 2010) on the blinded trial data:

$$\widehat{\beta}^{(cal)} = \arg\min_{\beta} \left[-\ell(\beta; \mathcal{D}_{trial}) + \lambda \sum_{j=1}^{c+1} \beta_j^2 \right],$$

where ℓ is the Cox partial log-likelihood and $\mathcal{D}_{trial}$ denotes the blinded trial data. The model includes the external score $\widehat{S}^{(ext)}$ and the c calibration covariates $W_{\mathcal{C}}$ as joint predictors. The ridge penalty λ shrinks all coefficients toward zero, with the optimal λ selected by 5-fold cross-validation maximizing the cross-validated partial likelihood. We implement the model using coordinate descent via the `glmnet` R package (Friedman et al., 2010) with $\alpha = 0$.

The calibrated score for patient i is:

$$\widehat{S}_i^{(cal)} = \widehat{\beta}_1 \widehat{S}_i^{(ext)} + \sum_{j \in \mathcal{C}} \widehat{\beta}_j W_{ij},$$

all estimated at the optimal λ selected by cross-validation.

Non-collapsibility in blinded calibration. Because the calibration model omits the treatment indicator A_i (the data is blinded), the Cox PH estimates $\widehat{\beta}^{(cal)}$ are subject to the non-collapsibility of the hazard ratio (Gail, Wieand, & Piantadosi, 1984). In linear regression, omitting a variable orthogonal to the included regressors — as randomized treatment A is to baseline covariates W — does not bias coefficient estimates. In the Cox model, however, omitting a prognostic treatment effect systematically attenuates the estimated coefficients of the included covariates toward zero, because the marginal hazard at a given covariate value is averaged over the unknown treatment assignment.

This attenuation is proportional to the magnitude of the treatment effect β_{trt} and the event rate. For $\beta_{trt} = \log 0.70$ and $N = 400$ (our primary setting), the attenuation is modest: the calibration coefficients are biased toward zero by approximately $\beta_{trt}^2/6 \approx 0.015$ in expectation (Struthers & Kalbfleisch, 1986). The ridge penalty λ partially compensates, since CV selects weaker shrinkage when the signal-to-noise ratio is reduced. Most importantly, the attenuation does not affect the validity of the primary analysis: under randomization, $\widehat{S}^{(cal)}$ remains a function of W only, and $A \perp W$ preserves asymptotic Type I error (Schuler et al., 2022, Theorem 1). Empirical verification is provided in Section 3.2 and a blinded-versus-unblinded calibration comparison is reported in Section 3.3.

2.4 Primary Analysis

The primary analysis uses the calibrated score as a covariate in a standard Cox PH model on the full, unblinded trial data:

$$\lambda(t \mid A_i, \widehat{S}_i^{(cal)}) = \lambda_0(t) \exp(\beta_{trt} A_i + \beta_{prog} \widehat{S}_i^{(cal)}),$$

with a robust sandwich variance estimator (Lin & Wei, 1989). The treatment effect is reported as $\exp(\widehat{\beta}_{trt})$ with a 95% confidence interval and Wald test p-value.

Validity. Because $\widehat{S}^{(cal)}$ is a function of W only, and $A \perp W$ by randomization, the Wald test for $H_0 : \beta_{trt} = 0$ preserves asymptotic Type I error regardless of how $\widehat{S}^{(cal)}$ was estimated (Schuler et al., 2022, Theorem 1). The sandwich variance estimator is consistent even when the score is estimated from the same data (Lin & Wei, 1989). Empirical verification is provided in our simulation study (Section 3).

2.5 Choosing the Calibration Set $\mathcal{C}$

The calibration set $\mathcal{C}$ should be pre-specified in the statistical analysis plan. We recommend selecting 3–8 covariates based on two criteria:

- **Clinical plausibility.** Covariates where the relationship with the outcome is likely to differ between the external and trial populations. Examples include acute-phase reactants (CRP, albumin) whose reference ranges shift with standard of care, biomarkers whose relevance varies by disease setting, and demographic factors with known cross-trial variability.
- **Prognostic strength.** Covariates with larger external coefficients should be prioritized, as correcting a strong coefficient has a larger impact on the score's predictive accuracy.

A data-driven sensitivity analysis can support the clinical pre-specification: compare the marginal distributions of each covariate between the external and trial populations (blinded, no unblinding needed). Covariates with large standardized mean differences $\Delta_j = |\bar{W}_j^{(trial)} - \bar{W}_j^{(ext)}|/\text{SD}(W_j^{(ext)})$ are candidates for $\mathcal{C}$.

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3. Simulation Study

3.1 Design

We simulate a two-arm, 1:1 randomized trial with progression-free survival as the primary endpoint. The data-generating process follows a Weibull proportional hazards model with shape parameter 1.5 and scale parameter 13 (baseline median survival approximately 10 months). Administrative censoring occurs at 24 months, with additional random dropout at 3% per year. Across scenarios, the expected number of events per arm ranges from 148 to 155 (mean total events 297–310 out of 400 patients).

Baseline covariates. Twenty baseline covariates are generated, all standardized: - 10 continuous (age, BMI, CRP, albumin, creatinine, WBC, hemoglobin, neutrophils, platelets, LDH) - 5 binary (sex, prior treatment, low eGFR, smoking, marker X) - 5 ordinal (ECOG, tumor stage, comorbidity index, symptom score, frailty)

The true prognostic model is a Cox PH with all 20 covariates, with coefficients chosen to yield a C-index of approximately 0.80 (LP standard deviation ≈ 1.3).

External data. An external dataset of $N_{ext} = 2000$ patients is generated from the same model as the trial, with two differences: - The baseline hazard may differ (external scale parameter varied by scenario) - The coefficients for the 5 calibration covariates $\mathcal{C} = \{\text{sex}, \text{marker\_x}, \text{CRP}, \text{albumin}, \text{LDH}\}$ may differ

The external model is a Cox PH fit on all 20 covariates, treated as a black box — only the predicted scores $\hat{S}^{(ext)}$ are used in the calibration step.

Scenarios.

Scenario	Description	$\Delta\beta$ on $\mathcal{C}$	N	β_{trt}
1. No shift	$\beta_{ext} = \beta_{trial}$ (null shift)	0	400	$\log 0.70$
2. Moderate shift	Small differences	$~$0.1–0.2 per coefficient	400	$\log 0.70$

Scenario	Description	$\Delta\beta$ on $\mathcal{C}$	N	β_{trt}
3. Severe shift	Marker X flips, sex becomes prognostic	$\Delta\beta \approx 0.3$–0.75	400	log 0.70
4. Treatment $\times$ covariate	Severe shift + marker X interacts with treatment	$\gamma = 0.5$	400	log 0.70
5. Null	Moderate shift, no treatment effect	$~$0.1–0.2	400	0
6. Non-PH	Severe shift, delayed onset (HR=1 for 0–2 months, then HR=0.70)	$\Delta\beta \approx 0.3$–0.75	400	log 0.70
7. Smaller effect	Severe shift, HR = 0.75	$\Delta\beta \approx 0.3$–0.75	400	log 0.75

Methods compared:

1. **Cox-2:** Cox PH model with 2 stratification variables (ECOG, sex). Represents conventional limited covariate adjustment.
2. **Full Model ("Oracle''):** Cox PH model with all 20 baseline covariates (matching the data-generating model). This is the oracle estimator — the best possible Cox model one could fit with unlimited trial data and no regulatory constraints on model complexity. It is not achievable in practice due to EPP limits, regulatory parsimony requirements, and missing data, but serves as a theoretical upper bound.

   Terminology note. We use `oracle'' throughout to refer to the DGP-matching model, standard in simulation literature. This usage differs from the causal inference convention where`oracle'' implies knowledge of unobserved potential outcomes.
3. **Stratified Log-Rank:** Non-parametric log-rank test stratified by ECOG and sex.
4. **PROCOVA:** Cox PH model with the external score $\widehat{S}^{(ext)}$ as sole covariate. Standard PROCOVA without calibration.
5. **Ridge-Cal (proposed):** Ridge-penalized Cox on blinded data with CV-selected λ (Section 2.3). Generates calibrated score $\widehat{S}^{(cal)}$ for primary analysis.
6. **MAP-Cox (sensitivity):** Robust MAP prior (Schmidli et al., 2014) borrowing external control data. Uses the precision-weighted approximation implemented in `map_proper()` to update calibration coefficients with commensurability-based scaling. MAP-Cox requires fitting a 21-parameter Cox model (all 20 covariates + treatment) on unblinded trial data, making it less parsimonious than Ridge-Cal (6 parameters). Results appear in Tables 1–2.

3.2 Results

Table 1: Empirical power (10,000 replicates per scenario).

Scenario	Std	Full Model	LR	PROCOVA	MAP-Cox	Ridge-Cal	Gain
:————————————	:—-:	:—-:	:–:	:——-:	:——-:	:——-:	:—-:
1. No shift	0.630	0.845	0.532	0.845	0.834	**0.837**	-0.008
2. Moderate	0.622	0.844	0.528	0.825	0.834	**0.834**	+0.009
3. Severe	**0.630**	**0.843**	**0.525**	**0.758**	**0.832**	**0.833**	**+0.075**
4. Interaction	0.631	0.865	0.530	0.730	0.856	**0.854**	+0.124
5. Null	0.055	0.065	0.051	0.053	0.059	**0.052**	—
6. Non-PH (2mo delay)	0.408	0.554	0.347	0.501	0.550	**0.551**	+0.050
7. Smaller effect (0.75)	0.456	0.682	0.371	0.572	0.665	**0.659**	**+0.087**

Note. The full model (20-covariate Cox matching the data-generating model) is the theoretical upper bound, not achievable in practice due to regulatory constraints on model complexity (FDA, 2023; EMA, 2015). Ridge-Cal uses only 6 parameters versus the full model's 21. The Non-PH scenario uses a 2-month delay, which reflects a realistic treatment onset lag.

Key observations. Under severe population shift (Scenario 3), Ridge-Cal recovers **+7.5 percentage points** of power over PROCOVA (0.758 to 0.833) and reduces bias by **80%** (0.035 to 0.007). The gain is larger under treatment-by-covariate interactions (+12.4 pp) and smaller effect sizes (+8.7 pp under HR = 0.75). Type I error is exactly nominal (0.052 under Scenario 5). The no-shift penalty is minimal ($-$0.8 pp). Under non-proportional hazards (Scenario 6, 2-month delayed effect), power is lower across all methods due to Cox model misspecification, but Ridge-Cal (0.551) nearly matches the full model (0.554) and beats PROCOVA (0.501) by +5.0 pp.

MAP-Cox achieves comparable power to Ridge-Cal across most scenarios, including 0.834 vs 0.837 (no shift) and 0.832 vs 0.833 (severe shift). However, MAP-Cox uses 21 parameters on unblinded data (all 20 covariates + treatment), whereas Ridge-Cal uses only 6 parameters on blinded data. Ridge-Cal thus matches or beats MAP-Cox despite being substantially more parsimonious and avoiding unblinding.

Table 2: Bias on the log-HR scale (10,000 replicates).

Scenario	PROCOVA bias	MAP-Cox bias	Ridge-Cal bias	Full model bias
1. No shift	0.001	-0.015	0.006	-0.015
2. Moderate	0.009	-0.014	0.006	-0.015
3. Severe	**0.035**	**-0.016**	**0.007**	**-0.016**
4. Interaction	0.046	-0.024	0.001	-0.024
5. Null	-0.001	-0.000	-0.001	-0.000
6. Non-PH	0.105	0.084	0.098	0.092
7. Smaller effect	0.030	-0.011	0.007	-0.011

Diagnostic C-index. Under severe shift, the C-index increases from 0.716 (score only) to 0.744 (score $+$ calibration covariates), correctly detecting miscalibration. Under no shift, the C-index is flat (0.741 vs 0.744), correctly indicating no calibration needed. The CV-selected ridge penalty is consistently $\lambda \approx 0.05$, providing moderate regularization across scenarios.

3.3 Sensitivity Analyses

Misspecified calibration set. To assess the penalty of including non-shifting covariates in $\mathcal{C}$, we augmented the calibration set with two strong prognostic covariates that do not shift between populations (age and hemoglobin). Under severe shift (2000 reps), the correct $\mathcal{C} = \{$sex, marker_x, CRP, albumin, LDH$\}$ yields Ridge-Cal power 0.826 and bias 0.007. The over-specified set $\mathcal{C}_{noise} = \mathcal{C} \cup \{age, hgb\}$ yields power 0.822 and bias 0.008 — a penalty of only 0.4 pp. Under the interaction scenario, the penalty is 0.8 pp (0.861 to 0.853). The ridge penalty effectively shrinks the non-shifting covariates toward zero, confirming that pre-specifying a slightly over-inclusive $\mathcal{C}$ is safe.

Lambda distribution. The CV-selected ridge penalty shows a tight distribution centered at $\lambda \approx 0.05$ (IQR 0.045–0.050) across all scenarios, with no extreme values indicating instability. The distribution is near-identical between correct and over-specified $\mathcal{C}$ settings, confirming robustness.

Blinded versus unblinded calibration. To quantify the non-collapsibility attenuation (Section 2.3), we compared the calibration coefficients from a blinded model (omitting A) against a model that includes A. The mean $\hat{\beta}_{\mathcal{C}}$ (S coefficient) differs by only 0.011–0.012 across all scenarios — consistent with the Struthers & Kalbfleisch (1986) approximation of $\beta_{trt}^2/6 \approx 0.015$. This confirms the attenuation is practically negligible for HR ≥ 0.70.

A proper MAP prior comparison (Schmidli et al., 2014) using precision-weighted updating of calibration coefficients is reported in Tables 1–2 above.

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4. Results

Tables 1 and 2 present the full simulation results (reproduced from Section 3.2). Ridge-Cal consistently outperforms PROCOVA under population shift, with power gains of +3.3 to +12.4 percentage points across scenarios. Bias on the log-HR scale is below 0.01 for Ridge-Cal under all non-Non-PH scenarios, compared to PROCOVA bias up to 0.046. Type I error is controlled at the nominal level for all methods (0.051–0.052). The no-shift penalty for Ridge-Cal is minimal ($-$0.8 pp).

MAP-Cox (Tables 1–2) achieves similar power to Ridge-Cal across most scenarios but uses 21 parameters on unblinded trial data (all 20 covariates + treatment) versus 6 parameters on blinded data. Ridge-Cal matches or exceeds MAP-Cox despite being substantially more parsimonious and requiring no unblinding.

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5. Discussion

5.1 Summary

We have proposed Ridge-Cal, a two-step procedure for diagnosing and correcting miscalibration of external prognostic scores using a trial's own blinded data. The method is inspired by parameter-efficient fine-tuning: the pre-trained score is frozen, and a small, regularized correction is learned for a pre-specified subset of covariates. The strength of the correction is selected automatically by cross-validation.

In simulations with 10,000 replicates under severe population shift, Ridge-Cal recovers 7.5 percentage points of power over standard PROCOVA (0.758 vs 0.833), reduces bias by 80% (0.035 vs 0.007), and maintains nominal Type I error (0.052 vs 0.053). The gain is larger under treatment-by-covariate interactions (+12.4 pp) and smaller effect sizes (+8.7 pp under HR = 0.75). When no shift is present, the power penalty is minimal ($-$0.8 pp) and the diagnostic correctly indicates no recalibration is needed.

5.2 Relationship to Existing Methods

Ridge-Cal complements PROCOVA rather than replacing it. PROCOVA provides the base score, and Ridge-Cal fine-tunes it when needed — analogous to how LoRA fine-tunes a pre-trained LLM rather than training from scratch.

Compared to Bayesian dynamic borrowing (Ibrahim & Chen, 2000; Hobbs et al., 2011; Schmidli et al., 2014), Ridge-Cal operates on the score rather than on patient-level data. It borrows prognostic structure (which covariates matter) rather than patient outcomes (which patients look similar). This distinction is critical when the external and trial populations differ qualitatively (e.g., different biomarker distributions) rather than quantitatively (e.g., different baseline hazards). A MAP prior comparison (Tables 1–2) confirms that Ridge-Cal achieves comparable or better power with far fewer parameters and blinded-data operation.

The ridge penalty makes Ridge-Cal more robust than naive recalibration (updating all coefficients freely), which overfits when the calibration sample is small. Cross-validated λ selection automates the regularization strength.

5.3 Limitations

Pre-specification of $\mathcal{C}$. The calibration set must be pre-specified in the statistical analysis plan. If a shifting covariate is left out, Ridge-Cal cannot correct it. Including non-shifting covariates adds noise but does not inflate Type I error. Pre-specification should be based on clinical judgment supported by blinded covariate distribution comparisons.

Linearity assumption. The additive correction assumes linear miscalibration. Non-linear patterns (e.g., effect changes only in a specific covariate range) would benefit from spline-based or random-forest-based calibration.

Bias from overfitting. In very small trials ($N < 200$), ridge may provide insufficient regularization. A fixed λ or stronger prior is recommended in such settings.

Non-collapsibility attenuation. As discussed in Section 2.3, blinded calibration attenuates the calibration coefficients due to the omitted treatment indicator. The attenuation is proportional to β_{trt}^2 and approximately $\beta_{trt}^2/6 \approx 0.015$ for HR $= 0.70$ (Struthers & Kalbfleisch, 1986). Our simulations confirm a mean attenuation of 0.011–0.012 in the calibration coefficients (Section 3.3), which is practically negligible. For larger treatment effects (HR < 0.50), the attenuation may be non-negligible, and a control-arm calibration (unblinding the control group only) should be considered as a sensitivity analysis.

Blinded calibration. The blinded variant assumes no strong treatment-by-covariate interactions. Our simulations show minimal impact, but the control-arm variant avoids this issue at the cost of partial unblinding.

No efficiency bound. We have not derived the semiparametric efficiency bound for the Ridge-Cal estimator.

5.4 Connection to LoRA and Future Directions

The ridge penalty λ plays an analogous role to the rank r in Low-Rank Adaptation (LoRA; Hu et al., 2022): $\lambda = 0$ permits unconstrained adaptation of the calibration coefficients, while $\lambda \to \infty$ recovers the base PROCOVA model. Cross-validated selection of λ provides automatic calibration strength tuning, similar to how the LoRA rank r is selected in practice. The key difference — LoRA constrains the dimension of the update, while ridge constrains its magnitude — reflects the different data regimes: LLMs have millions of parameters and abundant unlabeled data, while clinical trials have at most a few hundred events and require precise Type I error control. Ridge regularization is the natural choice for this setting.

Adapter-based calibration (NN-Cal). To assess whether a more flexible calibration is beneficial, we implemented a small neural network adapter (6–3–1 architecture with ReLU activations and ridge-regularized weights, trained via Adam on the Cox partial likelihood). Across six non-linear scenarios (threshold effects, interactions, U-shaped hazards, and combinations), Ridge-Cal matched or exceeded the neural network in all cases (200 reps each). The linear correction is sufficient at $N = 400$ because the external score captures most of the prognostic structure; the calibration need only adjust coefficients, not learn new representations. A neural network adapter may offer advantages in larger trials ($N > 800$) with more complex non-linear miscalibration patterns.

Meta-learning. With multiple historical trials, a meta-learned prior on calibration coefficients could automate $\mathcal{C}$ selection.

Online calibration. In adaptive designs, the calibration could update at each interim using accumulating blinded data, with λ scheduled to decrease over time.

5.5 Conclusion

Ridge-Cal addresses a specific limitation of PROCOVA: the assumption that external prognostic scores remain well-calibrated for the trial population. By applying a ridge-penalized Cox model to blinded trial data, Ridge-Cal diagnoses and corrects miscalibration with respect to a pre-specified set of covariates. The ridge penalty protects against overfitting when the calibration sample is small, and cross-validated selection automates the regularization strength. In 10,000-rep simulations, Ridge-Cal improves power over PROCOVA under all forms of population shift by 3.3 to 12.4 percentage points with exact Type I error control and a minimal no-shift penalty (0.8 pp). A MAP-Cox comparison (Tables 1–2) confirms comparable or better power with far fewer parameters and blinded-data operation. We recommend Ridge-Cal as a sensitivity analysis in any PROCOVA-qualified trial.

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